

DIGITAL SYSTEMS

LECTURE 15 & 16

MESSAGES & SCHEDULING

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Hilary Term 2026



Department of
COMPUTER
SCIENCE

Following the course of past years by Mike Spivey

C Pointer Syntax

- Variable containing 1

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int a = 1;
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- Variable able to contain address to memory location containing int

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- Set c to contents of a

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int c = *b
```

C Pointer Syntax for Structs

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typedef struct dst {  
    int feet;  
    float inch;  
} distance;
```

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- Struct-typed variable

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distance a; a.feet = 9001; a.inch = 11;
```

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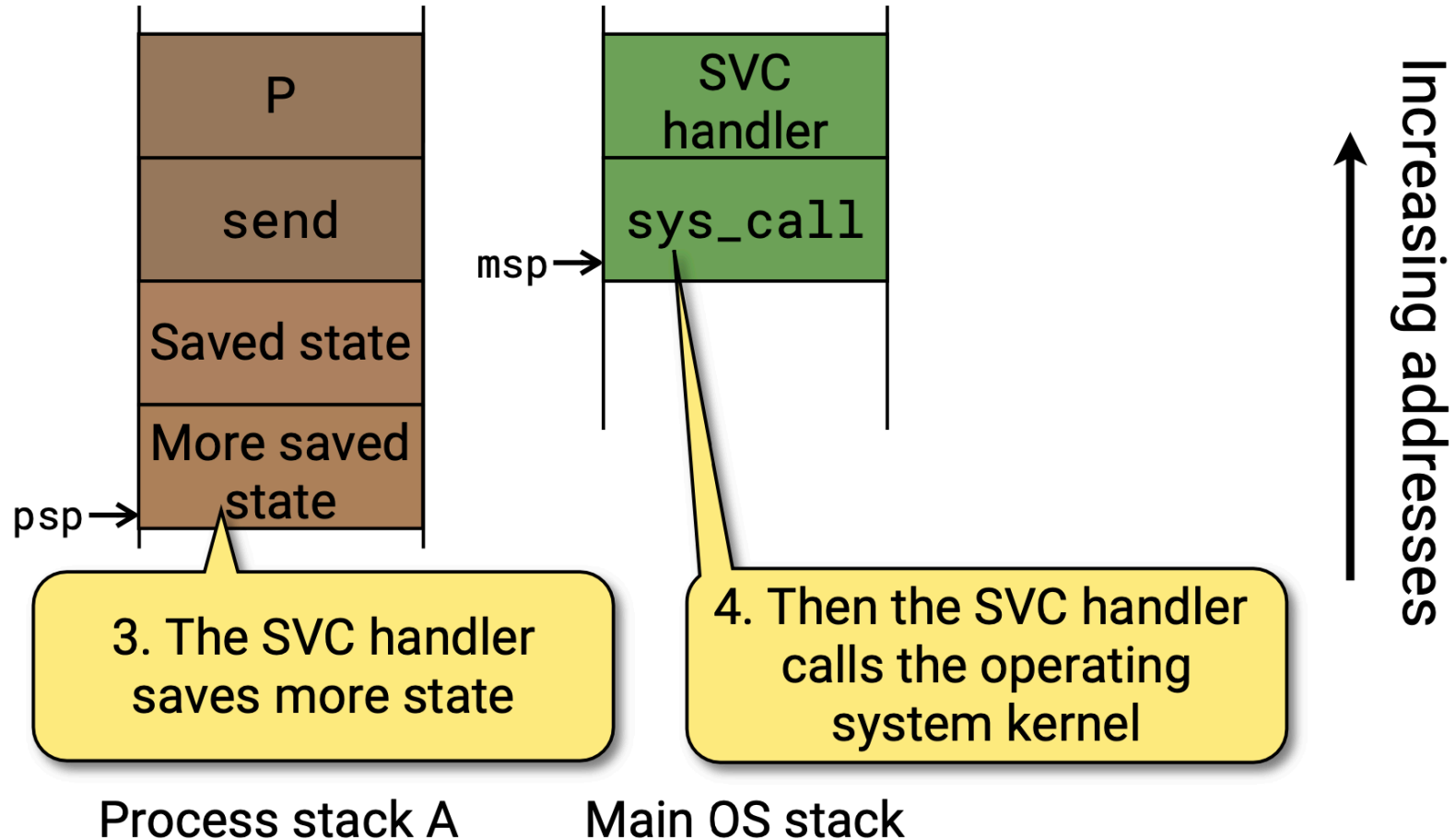
- Struct-typed variable
distance a; a.feet = 9001; a.inch = 11;
- Pointer to distance-typed variable, assign address of a
distance* b; b = &a;
- Content of address in b
(*b).inch == a.inch; /* Or... */ b->inch == a.inch;

Today

? How are processes started?

? What happens to the Operating System on startup?

Reminder: State Saved on Stack



Starting a Process

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- All other registers =

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- `lr` = address of exit stub, in case body returns.
- All other registers = Doesn't matter!

Our Program

```
void init(void) {  
    serial_init();  
    timer_init();  
    start("Heart", heart_task, 0, STACK);  
    start("Prime", prime_task, 0, STACK);  
}
```

Discuss in code... ex-heart.c

Creating “fake” Stack Frame

```
int start(char *name, void (*body)(int), int arg, int stksize){
    ...
    unsigned *sp = p->sp - FRAME_WORDS;
    memset(sp, 0, 4*FRAME_WORDS);
    // Macros for offsets in stack, ***_SAVE
    sp[PSR_SAVE] = INIT_PSR; // Thumb bit switched on
    sp[PC_SAVE] = (unsigned) body & ~0x1; // Activate the proc
    sp[LR_SAVE] = (unsigned) exit; // Make it return to exit()
    sp[R0_SAVE] = (unsigned) arg; // Pass arg in r0
    sp[ERV_SAVE] = MAGIC;
    p->sp = sp;
    ... }
```

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At this point, every process has a stack frame that:

- will restore the pc at the start of the process's function (i.e. it will run the function)
- has values for all state and registers that respects the restoring convention (manual part + hardware part),
- has values in the registers that respect the calling convention of C functions (e.g. integer argument in r0).

Starting the OS

After `init()` called, all tasks have stack frames that will allow a context switch into them. (Introduce `os_current` variable.)

```
void __start(void) {  
    /* Create idle process */  
    ...  
    /* Call the application's setup function */  
    init();  
    /* The main program morphs into the idle process. */  
    os_current = idle_proc;  
    set_stack(os_current->sp);  
    idle_task();  
}
```


Idle Process

```
void idle_task(void) {  
    /* Pick a genuine process to run */  
    yield();  
  
    /* When there's nothing to do: */  
    while (1) pause();  
}
```

Where are we now?

We understand *how* tasks are switched and started.

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(Q: remind me how)
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We understand *how* tasks are switched and started.

- OS is an interrupt that performs context switches
(Q: remind me how)
- During context switch, interrupt stores some state on stack, assembly code stores the rest.

What is left to understand?

? How does the OS keep track of processes?

We have already seen that we need to start processes, and keep track of various process-specific quantities (e.g. stack location), in order to do context switches. How does the OS do this?

? How does the OS decide which process to run next?

? How are messages transferred between processes?

And in particular, how is this done in interrupts?

? How does the OS keep track of processes?

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Process Structure

```
struct _proc {  
    int pid;           // Process ID  
    char name[16];     // Name for debugging  
    unsigned state;    // SENDING, RECEIVING, etc.  
    unsigned *sp;      // Saved stack pointer  
    int priority;      // Priority: 0 is highest  
    proc waiting;      // Processes waiting to send  
    int pending;       // Whether interrupt pending  
    int msgtype;       // Message to send or receive  
    message *msgbuf;   // Pointer to message buffer  
    proc next;         // Next process in ready/send queue  
};
```

Possible states

- DEAD: process that has exited. It is also the value present in slots of the table that have never been used.
- ACTIVE: process is ready to run. The process is either the current process, or the ready queue waiting to become current.
- SENDING: process is waiting to send to another; the process will appear in the sending queue of the destination.
- RECEIVING: process is waiting to receive a message; the msgtype field will specify who may send to it.
- SENDREC: call to sendrec, combining the functions of send and receive.
- IDLING: idle process, which runs when no other process is ready.

Process Structure

```
typedef struct _proc *proc;  
...  
static proc os_ptable[NPROCS];
```

In start(...):

```
    proc p = create_proc(name, roundup(stksize, 8));
```

In create_proc(...) we

- Initialise all the fields to sensible defaults / based on arguments.
- Find a memory location to initialise the stack (sbrk(...))
- Find a memory location for proc variable (new_proc(...))

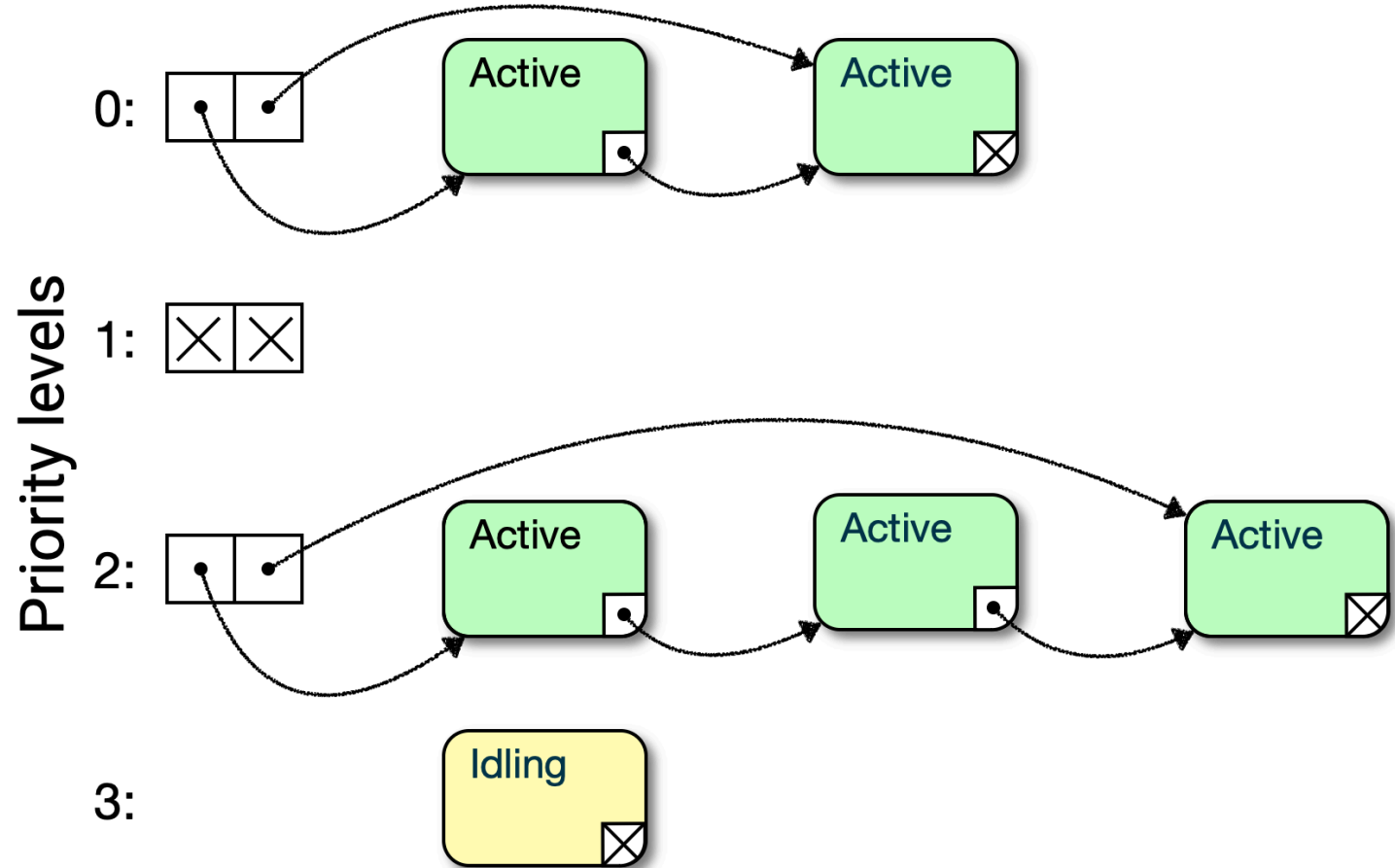
Draw memory layout (heap, stacks, ..., processes, main stack). ⇐

? How does the OS decide which process to run next?

Start with the easy case: context switch initiated by `yield()`.

Ready Queues

In ACTIVE processes,
the ready queue is
implemented using
the next field.



Ready Queues

```
#define NPRI0 3          /* Number of non-idle priorities */  
  
...  
static struct _queue {  
    proc head, tail;  
} os_readyq[NPRI0];
```

Last line of `start(...)`, after setting up fake exception frame...

```
make_ready(p);
```

Adds process to appropriate queue.

Adding to the Queue: make_ready()

```
static inline void make_ready(proc p) {  
    int prio = p->priority;  
    if (prio == P_IDLE) return;  
  
    p->state = ACTIVE;  
    p->next = NULL;    // Because last in queue  
    queue q = &os_readyq[prio];  
    if (q->head == NULL)  
        q->head = p;  
    else  
        q->tail->next = p;  
    q->tail = p;      }  
}
```

Full Story of Startup

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- Creates handcrafted (rather than hardware) stack frame, so process can be started through normal return.
- Calls `make_ready()` to add process to appropriate queue.
At initialisation, all processes are ACTIVE and in the ready queue.
- Make current function the idle task.
- `yield()`, so OS starts scheduling procedure.

Selecting Next Process

We end up in `system_call(...)` again.

```
unsigned *system_call(unsigned *psp) {  
    short *pc = (short *) psp[PC_SAVE]; /* Program counter */  
    int op = pc[-1] & 0xff; // Syscall number from SVC instr  
    os_current->sp = psp;    // Save sp of the current process  
    ...  
    switch (op) {  
    case SYS_YIELD:  
        // Add current to back of Q. Choose next process.  
        make_ready(os_current); choose_proc(); break;  
        ... }  
    return os_current->sp; } // Return sp for next process
```

Selecting Next Process

Find highest priority process to run, remove it from queue, return.

```
static inline void choose_proc(void) {  
    for (int p = 0; p < NPRI0; p++) {  
        queue q = &os_readyq[p];  
        if (q->head != NULL) {  
            os_current = q->head;  
            q->head = os_current->next;  
            return;  
        }  
    }  
    os_current = idle_proc;  
}
```


System Call after send()

Just like yield(), send() simply causes a syscall interrupt.

```
unsigned *system_call(unsigned *psp) {  
    short *pc = (short *) psp[PC_SAVE];  
    int op = pc[-1] & 0xff;  
    os_current->sp = psp;  
    switch (op) {  
    case SYS_SEND:  
        mini_send(arg(0, int), arg(1, int), arg(2, message *));  
        break; ...  
    }  
    return os_current->sp;  
}
```

Sending Messages

Remember the rules for communication by messages!

- Transfer message to destination. Destination READY. Add to queue.
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Sending Messages

Remember the rules for communication by messages!

- Transfer message to destination. Destination READY. Add to queue.
- Multiple possible reasons for this (Q: which?)
- Must (somehow) keep sender in queue until process is ready to receive it.

Sending Queues

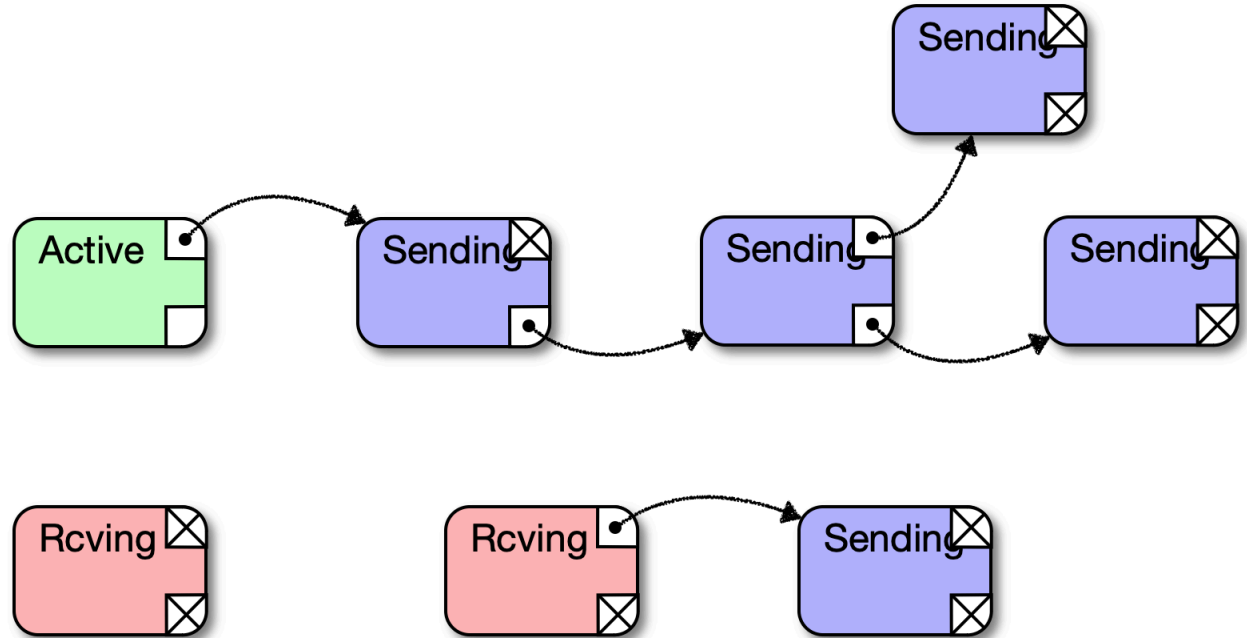
Processes can have multiple other processes waiting to send.

Next process trying to send is stored in waiting.

Rest of the send queue is chained through next.

Q: Does this conflict with the ready queue?

Each process has a queue of others waiting to send to it.



- Active process has 3 processes waiting to send to it.
- One process waiting to send also has a process waiting to send to *it*.
- One waiting to receive, but nothing to send to it yet.
- One waiting to receive, with one waiting to send to it, but not the right type of message.

System call after send()

```
static void mini_send(int dest, int type, message *msg) {  
    int src = os_current->pid;  
    proc pdest = os_ptable[dest];  
    if (accept(pdest, type)) { // Receiver is waiting  
        deliver(pdest->msgbuf, src, msg);  
        make_ready(pdest); make_ready(os_current); // *DISCUSS*  
    } else {  
        os_current->state = SENDING;  
        queue_send(pdest); // Sender waits. Join receiver's queue  
    }  
    choose_proc();  
}
```

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But this time, if we **receive** a message.

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Receiving Messages

```
static void mini_receive(int type, message *msg) {  
    if (/* interrupt is due */) { // later...  
    } else {  
        proc psrc = find_sender(os_current, type); // search send Q  
        if (psrc != NULL) { // Is a sender waiting?  
            deliver(msg, psrc->pid, psrc->msgbuf);  
            make_ready(os_current); make_ready(psrc);  
        } else { // No luck: we must wait  
            set_state(os_current, RECEIVING, type, msg);  
        }  
    }  
    choose_proc();  
}
```

? How are messages transferred between processes?

And in particular, how is this done in interrupts?

Interrupt Handler

```
void default_handler(void) {  
    int irq = active_irq(), task;  
    if (irq < 0 || (task = os_handler[irq]) == 0)  
        panic("Unexpected interrupt %d", irq);  
    disable_irq(irq);  
    interrupt(task);  
}
```

Discuss code.

Summary

- Process table
- Process states
- Ready queues
- Send queues
- Implementation of delivery schedule of messages

Notes contain another example of a device driver, and a discussion of SENDREC. Do take a look at this.

End of Term!

Next term, start from the complete bottom of the stack:

? How to we build a computer?

... starting from a transistor